Mathematical Statistics Final Cheat Sheet

Weeks 9-14: Functions of RVs, Sampling, Estimation & Testing

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1. Functions of Random Variables

Method 1: CDF Method (Univariate)
For $Y = g(X)$:

1. Find CDF: $F_Y(y) = P(Y \le y) = P(g(X) \le y)$
2. Differentiate: $f_Y(y) = F_Y'(y)$

Method 2: Transformation Formula
If $Y = g(X)$ is monotonic and differentiable:

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{dx}{dy}\right| = f_X(x)\left|\frac{dx}{dy}\right|_{x=g^{-1}(y)}$$

For many-to-one (e.g., $Y = X^2$), sum over roots:

$$f_Y(y) = \sum_i f_X(x_i(y))\left|\frac{dx_i}{dy}\right|$$

Method 3: MGF Method
If $M_Y(t) = M_Z(t)$, then $Y \overset{d}{=} Z$ (same distribution).
Method 4: Jacobian (Bivariate)
For $(U, V) = g(X, Y)$ with inverse $(X, Y) = h(U, V)$:

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v))|J|$$

Jacobian: $J = \det\left[\frac{\partial(x,y)}{\partial(u,v)}\right]$
Convolution (Sum of Independent RVs)
If X, Y independent, $Z = X + Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)dx$$

Discrete: $p_Z(z) = \sum_x p_X(x)p_Y(z - x)$
Order Statistics
For i.i.d. $X_1, \dots, X_n$ with CDF F and PDF f:
Minimum: $X_{(1)} = \min\{X_1, \dots, X_n\}$

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$$

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1}f(x)$$

Maximum: $X_{(n)} = \max\{X_1, \dots, X_n\}$

$$F_{X_{(n)}}(x) = [F(x)]^n$$

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1}f(x)$$

k-th order statistic:

$$f_{X_{(k)}}(x) \propto [F(x)]^{k-1}[1 - F(x)]^{n-k}f(x)$$

Uniform$(0, 1)$: $X_{(k)} \sim \text{Beta}(k, n - k + 1)$
Key Transformation Results
If $Z \sim N(0, 1)$:

- $Z^2 \sim \chi^2(1)$
- $\sum_{i=1}^n Z_i^2 \sim \chi^2(n)$ (independent Z_i)

If $X \sim \text{Uniform}(0, 1)$:

- $Y = -\log(X) \sim \text{Exp}(1)$

2. Sampling Distributions

Law of Large Numbers (LLN)
If $X_1, X_2, \dots$ i.i.d. with $\mathbb{E}[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2 < \infty$:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{P} \mu$$

Chebyshev proof: $P(|\bar{X}_n - \mu| > \varepsilon) \le \frac{\sigma^2/n}{\varepsilon^2} \to 0$
Central Limit Theorem (CLT)
If X_i i.i.d. with mean μ and variance $\sigma^2 > 0$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

Equivalently: $\bar{X}_n \approx N(\mu, \sigma^2/n)$ for large n
For sums: $S_n = \sum X_i \approx N(n\mu, n\sigma^2)$
Continuity Correction (Discrete → Continuous)
For $X \sim B(n, p)$: $P(X \le k) \approx P\left(Z \le \frac{k+0.5-np}{\sqrt{np(1-p)}}\right)$
Rule: Normal approx reliable if $np(1 - p) \gtrsim 10$
Chi-Square Distribution: $\chi^2(k)$
If $Z_1, \dots, Z_k \sim N(0, 1)$ independent:

$$\sum_{i=1}^k Z_i^2 \sim \chi^2(k)$$

PDF: $f(x) = \frac{1}{2^{k/2}\Gamma(k/2)}x^{k/2-1}e^{-x/2}$, $x > 0$
$\mathbb{E}[\chi^2(k)] = k$, $\text{Var}(\chi^2(k)) = 2k$
MGF: $M(t) = (1 - 2t)^{-k/2}$, $t < 1/2$
Additivity: If $X \sim \chi^2(k_1)$, $Y \sim \chi^2(k_2)$ independent:

$$X + Y \sim \chi^2(k_1 + k_2)$$

t-Distribution: $t(n)$
If $Z \sim N(0, 1)$, $V \sim \chi^2(n)$ independent:

$$T = \frac{Z}{\sqrt{V/n}} \sim t(n)$$

$\mathbb{E}[T] = 0$ $(n > 1)$, $\text{Var}(T) = \frac{n}{n-2}$ $(n > 2)$
As $n \to \infty$: $t(n) \to N(0, 1)$
F-Distribution: $F(n_1, n_2)$
If $V_1 \sim \chi^2(n_1)$, $V_2 \sim \chi^2(n_2)$ independent:

$$F = \frac{V_1/n_1}{V_2/n_2} \sim F(n_1, n_2)$$

$\mathbb{E}[F] = \frac{n_2}{n_2-2}$ $(n_2 > 2)$
If $F \sim F(n_1, n_2)$: $\frac{1}{F} \sim F(n_2, n_1)$
If $T \sim t(n)$: $T^2 \sim F(1, n)$
Normal Sample: Key Results
If $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ i.i.d.:

- $\bar{X} \sim N(\mu, \sigma^2/n)$
- $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n - 1)$ where $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$
- $\bar{X}$ and S^2 are independent
- $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ (σ known)
- $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n - 1)$ (σ unknown)

Delta Method
If $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, \tau^2)$ and $g'(\theta) \ne 0$:

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \Rightarrow N(0, [g'(\theta)]^2\tau^2)$$

Variance: $\text{Var}(g(\hat{\theta}_n)) \approx [g'(\theta)]^2\text{Var}(\hat{\theta}_n)$

3. Point Estimation

Bias and MSE
Bias: $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$
Unbiased: $\text{Bias}(\hat{\theta}) = 0$
Variance: $\text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$
MSE: $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$
Consistency
$\hat{\theta}_n$ is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \to \infty$
Sufficient condition: If unbiased and $\lim_{n\to\infty}\text{Var}(\hat{\theta}_n) = 0$
Or: If $\lim_{n\to\infty}\text{MSE}(\hat{\theta}_n) = 0$
Relative Efficiency
For two unbiased estimators $\hat{\theta}_1$, $\hat{\theta}_2$:

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

If eff > 1: $\hat{\theta}_1$ is more efficient (smaller variance)
Method of Moments (MoM)
Equate sample moments to population moments:

1. $\mu_j' = \mathbb{E}[X^j]$ (population moments)
2. $m_j = \frac{1}{n}\sum X_i^j$ (sample moments)
3. Solve: $m_j = \mu_j'(\theta)$ for θ

Properties: Simple, consistent, but may not be efficient
Maximum Likelihood (MLE)
Likelihood: $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$
Log-likelihood: $\ell(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$
MLE: $\hat{\theta}_{\text{MLE}} = \arg\max_\theta \ell(\theta)$
Finding MLE: Solve $\frac{d\ell}{d\theta} = 0$ (score equation)
MLE Properties:

- Invariance: If $\hat{\theta}$ is MLE of θ, then $g(\hat{\theta})$ is MLE of $g(\theta)$
- Consistent under regularity conditions
- Asymptotically efficient
- Asymptotic normality: $\hat{\theta}_n \approx N(\theta, \frac{1}{nI(\theta)})$

Fisher Information
Single observation:

$$I(\theta) = \mathbb{E}\left[\left(\frac{\partial \log f(X; \theta)}{\partial\theta}\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2 \log f(X; \theta)}{\partial\theta^2}\right]$$

For i.i.d. sample: $I_n(\theta) = n \cdot I(\theta)$
Cramér-Rao Lower Bound (CRLB)
For any unbiased estimator $\hat{\theta}$:

$$\text{Var}(\hat{\theta}) \ge \frac{1}{I_n(\theta)} = \frac{1}{nI(\theta)}$$

Efficiency: $\text{eff}(\hat{\theta}) = \frac{1/(nI(\theta))}{\text{Var}(\hat{\theta})} \in [0, 1]$
Estimator achieving bound is called efficient.
Common Fisher Information:

- Bernoulli(p): $I(p) = \frac{1}{p(1-p)}$
- Poisson(λ): $I(\lambda) = \frac{1}{\lambda}$
- Normal(μ, σ^2) (known σ): $I(\mu) = \frac{1}{\sigma^2}$
- Exponential(λ): $I(\lambda) = \frac{1}{\lambda^2}$

Sufficiency
$T(X)$ is sufficient for θ if conditional distribution of X given T doesn't depend on θ.
Factorization Theorem:
T is sufficient iff likelihood factors as:

$$L(\theta; x) = g(T(x), \theta) \cdot h(x)$$

where g depends on data only through T, h doesn't depend on θ.
Examples:

- Normal(μ, σ^2) (known σ): $\bar{X}$ is sufficient for μ
- Poisson(λ): $\sum X_i$ is sufficient for λ
- Exponential(λ): $\sum X_i$ is sufficient for λ

Completeness & UMVUE
T is complete if $\mathbb{E}[g(T)] = 0$ for all θ implies $P(g(T) = 0) = 1$.
Lehmann-Scheffé Theorem:
If T is complete sufficient and $\phi(T)$ is unbiased for $\tau(\theta)$, then $\phi(T)$ is UMVUE (unique minimum variance unbiased estimator).

4. Confidence Intervals

Definition
$(1 - \alpha)100\%$ CI for θ: Random interval $[L, U]$ with

$$P(L \le \theta \le U) = 1 - \alpha$$

Interpretation: $(1 - \alpha)100\%$ of intervals contain true θ.
Common CIs
Mean (σ known, large n):

$$\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Mean (σ unknown, Normal):

$$\bar{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

Proportion (large n):

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Variance (Normal):

$$\frac{(n - 1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n - 1)s^2}{\chi^2_{1-\alpha/2,n-1}}$$

Ratio of Variances:

$$\frac{s_1^2/s_2^2}{F_{\alpha/2,n_1-1,n_2-1}} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{s_1^2}{s_2^2} \cdot F_{\alpha/2,n_1-1,n_2-1}$$

Two-Sample Mean (equal var):

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $s_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$
Sample Size Determination
For mean with margin E and confidence $(1-\alpha)$:

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

For proportion:

$$n = \hat{p}(1-\hat{p})\left(\frac{z_{\alpha/2}}{E}\right)^2$$

Conservative (no prior $\hat{p}$): Use $\hat{p} = 0.5$ (gives largest n)

5. Hypothesis Testing

Framework
Null hypothesis H_0: Status quo (assume true)
Alternative H_1: What we want evidence for
Types:

- Two-sided: $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$
- Upper: $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$
- Lower: $H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$

Error Types
Type I Error (False Positive):
$\alpha = P(\text{Reject } H_0 | H_0 \text{ true})$
Type II Error (False Negative):
$\beta = P(\text{Fail reject } H_0 | H_1 \text{ true})$
Power: $1 - \beta = P(\text{Reject } H_0 | H_1 \text{ true})$
p-Value
Probability under H_0 of observing test statistic as extreme or more extreme.
Decision: Reject H_0 if $p \leq \alpha$
Interpretation:

- $p \leq 0.01$: Very strong evidence against H_0
- $0.01 < p \leq 0.05$: Moderate evidence
- $0.05 < p \leq 0.10$: Weak evidence
- $p > 0.10$: Little/no evidence

Common Test Statistics
Z-test (mean, σ known):

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \text{ under } H_0$$

t-test (mean, σ unknown):

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0$$

Proportion test (large n):

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0,1) \text{ under } H_0$$

Chi-square (variance):

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi^2_{n-1} \text{ under } H_0$$

Two-sample t-test (equal var):

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p\sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2} \text{ under } H_0$$

F-test (variance ratio):

$$F = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1} \text{ under } H_0 : \sigma_1^2 = \sigma_2^2$$

Rejection Regions
Two-sided:

- Z or t: $|T| > t_{\alpha/2}$
- χ^2: $\chi^2 < \chi^2_{1-\alpha/2}$ or $\chi^2 > \chi^2_{\alpha/2}$
- F: $F < F_{1-\alpha/2}$ or $F > F_{\alpha/2}$

One-sided (upper):

- Z or t: $T > t_\alpha$
- χ^2: $\chi^2 > \chi^2_\alpha$
- F: $F > F_\alpha$

Power & Sample Size
Power: $\pi(\mu_1) = \Phi\left(\frac{\mu_1-\mu_0}{\sigma/\sqrt{n}} - z_\alpha\right)$
Sample size: $n = \left(\frac{(z_\alpha+z_\beta)\sigma}{\mu_1-\mu_0}\right)^2$
Two-sided: Use $z_{\alpha/2}$ instead of z_α
CI-Testing Duality
If $(1-\alpha)$ CI for θ is $[L, U]$:

- $\theta_0 \notin [L, U]$: Reject $H_0 : \theta = \theta_0$ at level α
- $\theta_0 \in [L, U]$: Fail to reject H_0

Multiple Testing
Bonferroni correction for m tests with FWER α:
Test each at level α/m
Family-wise error rate without correction: $\approx m\alpha$

6. Likelihood Ratio Tests

Neyman-Pearson Lemma
For simple $H_0 : \theta = \theta_0$ vs simple $H_1 : \theta = \theta_1$:
Most powerful test of size α rejects when:

$$\Lambda(x) = \frac{L(\theta_0; x)}{L(\theta_1; x)} \leq k$$

where k satisfies $P(\Lambda \leq k | H_0) = \alpha$
General LRT
For composite hypotheses $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$:
Likelihood ratio:

$$\lambda(x) = \frac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)} = \frac{L(\hat{\theta}_0; x)}{L(\hat{\theta}; x)}$$

where $\hat{\theta}_0$ = MLE under H_0, $\hat{\theta}$ = unrestricted MLE
Properties: $0 \leq \lambda(x) \leq 1$, smaller $\lambda \rightarrow$ stronger evidence against H_0
Wilks' Theorem
Under H_0 and regularity conditions:

$$-2\log\lambda(X) \xrightarrow{d} \chi^2_\nu$$

where $\nu = \dim(\Theta) - \dim(\Theta_0)$ (df = difference in parameters)
Rejection: $-2\log\lambda(x) > \chi^2_{\alpha,\nu}$
Three Asymptotic Tests
For $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$:
1. Wald Test:

$$W = n(\hat{\theta} - \theta_0)^2 I(\hat{\theta}) \xrightarrow{d} \chi^2_1$$

Based on MLE and asymptotic normality.

2. Score Test (Rao's):

$$\text{Score} = \frac{[S(\theta_0)]^2}{nI(\theta_0)} \xrightarrow{d} \chi^2_1$$

where $S(\theta_0) = \left.\frac{\partial \log L}{\partial \theta}\right|_{\theta=\theta_0}$
Based on score evaluated at θ_0.
3. LRT:

$$\text{LRT} = -2\log\lambda = 2[\ell(\hat{\theta}) - \ell(\theta_0)] \xrightarrow{d} \chi^2_1$$

All three asymptotically equivalent. Ordering: Score $\leq$ LRT $\leq$ Wald
Goodness-of-Fit
Chi-square test:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \xrightarrow{d} \chi^2_{k-1-p}$$

O_i = observed, E_i = expected, p = params

7. Key Distributions Summary

Dist	Mean	Var	Support
$\chi^2(k)$	k	$2k$	$(0, \infty)$
$t(n)$	0	$\frac{n}{n-2}$	$(-\infty, \infty)$
$F(n_1, n_2)$	$\frac{n_2}{n_2-2}$	-	$(0, \infty)$

Relationships:

- $\sum_{i=1}^k Z_i^2 \sim \chi^2(k)$, $Z_i \sim N(0,1)$ iid
- $\frac{Z}{\sqrt{V/n}} \sim t(n)$, $Z \sim N(0,1)$, $V \sim \chi^2(n)$
- $\frac{V_1/n_1}{V_2/n_2} \sim F(n_1, n_2)$, $V_i \sim \chi^2(n_i)$
- $T^2 \sim F(1, n)$ if $T \sim t(n)$
- $\chi^2(k_1) + \chi^2(k_2) = \chi^2(k_1 + k_2)$ (independent)

8. Important Formulas

Sample Statistics (Normal):

$$\bar{X} \sim N(\mu, \sigma^2/n)$$
$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$
$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

Two Samples (Normal, equal var):

$$\frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$$

Asymptotic Normality:

$$\hat{\theta}_n \approx N\left(\theta, \frac{1}{nI(\theta)}\right)$$

Delta Method:

$$g(\hat{\theta}_n) \approx N\left(g(\theta), \frac{[g'(\theta)]^2}{nI(\theta)}\right)$$

LRT Asymptotic:

$$-2\log\lambda \xrightarrow{d} \chi^2_\nu$$